



The Better Policy Project

The Prudent Risk Management Approach to Macroeconomic Policy:

Calibration Approach

The Better Policy Project Team
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Housekeeping

- What should be in the Phillips curve?
 - Real exchange rate gap or the change in the real exchange rate gap?
- Inflation Targeting is about missing targets
 - Example of New Zealand in the 1990s



In India and Armenia , The FPAS Mark 2 Framework

• Standard QPM

- Clean
- Tractable
- Organizing device

• Endocred QPM

- If you turn off credibility
- If you turn off the non-linearities
- Results will look like the standard QPM

• Auxiliary models

- Fiscal policy
- Endogenous money creation (tomorrow)
- Financial-linkages
- BOP, trade
- Asset prices



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General Guidelines: Berg, Karam Laxton (2006)

- “We take an eclectic approach to parameterizing the model.
 - “The proposed parameter values in this section are based on experience with this type of model in a number of countries. Many have been implemented at central banks and are unpublished.”
1. Does the calibration fit the historical data/narrative? – QPM models with the Kalman Filter
 2. Experience producing realistic scenarios (guiding principle in EMs with poor data) – calibrate for the future not the past
 3. Big pool of estimated/calibrated country models to help compare against



The Output Gap Equation

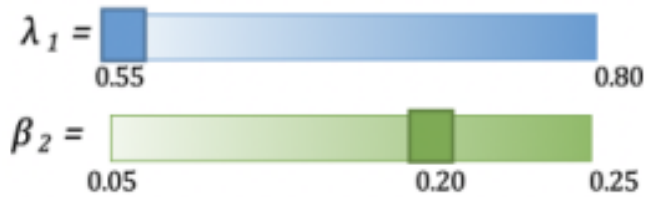
$$\hat{y}_t = \beta_1 * \hat{y}_{t-1} + \beta_2 * \hat{y}_{t+1} - \beta_3 * \hat{r}_{t-1} + \beta_4 * \hat{\zeta}_{t-1} + \beta_5 * \hat{y}_t^w + \varepsilon_t^{\hat{y}}$$





The Phillips Curve

$$\pi_t = \lambda_1 \pi_{t+4}^e + (1 - \lambda_1) \pi_{t-1} + \lambda_2 \left(\frac{\hat{y}_{t-1}}{\hat{y}_{max} - \hat{y}_{t-1}} \hat{y}_{max} \right) + \beta_1 * (\hat{z}_t - \hat{z}_{t-1}) + \beta_2 * \hat{z}_t + \varepsilon_t^\pi \quad (9)$$

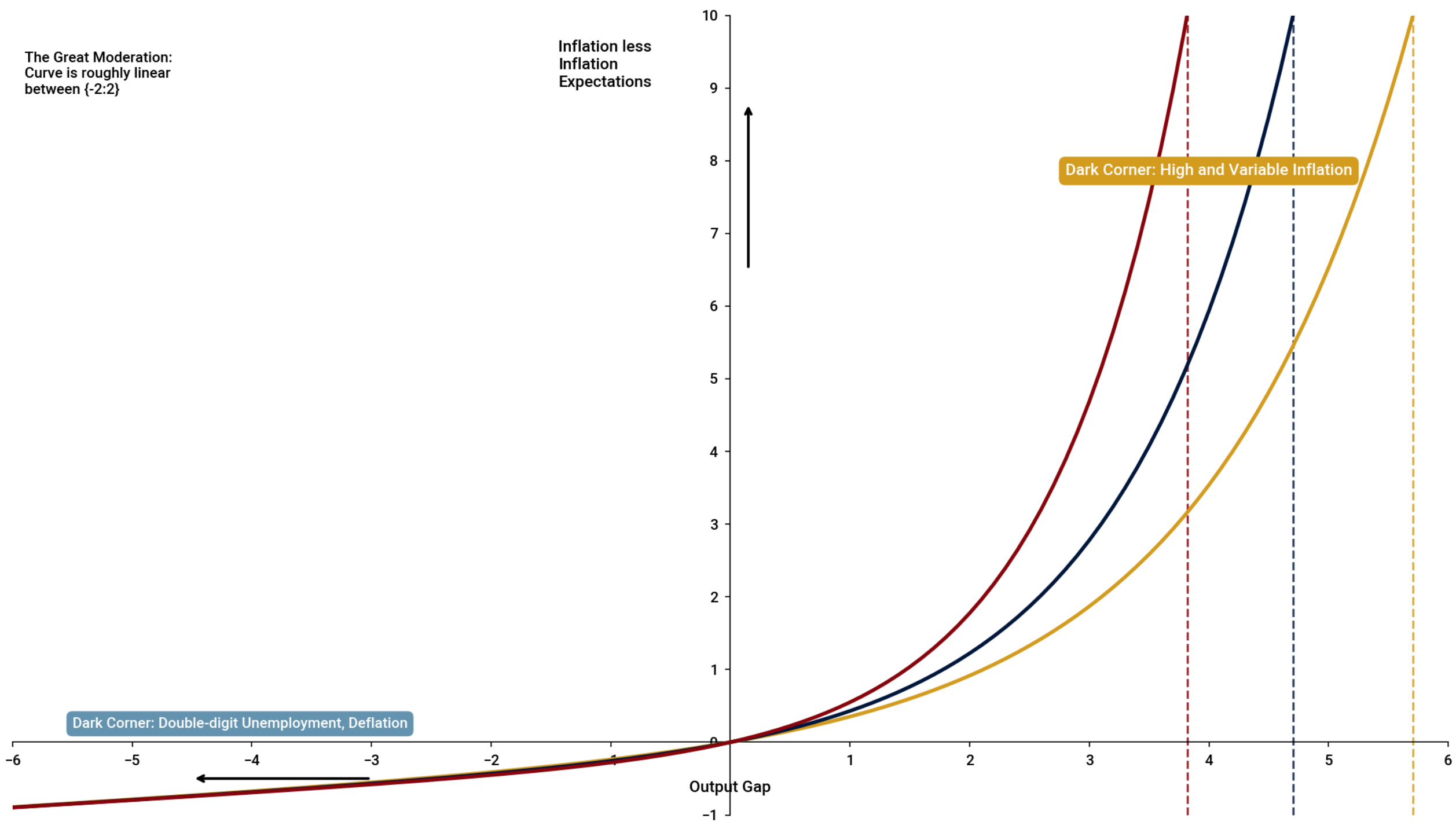


The Great Moderation:
Curve is roughly linear
between $\{-2:2\}$

Inflation less
Inflation
Expectations

Dark Corner: High and Variable Inflation

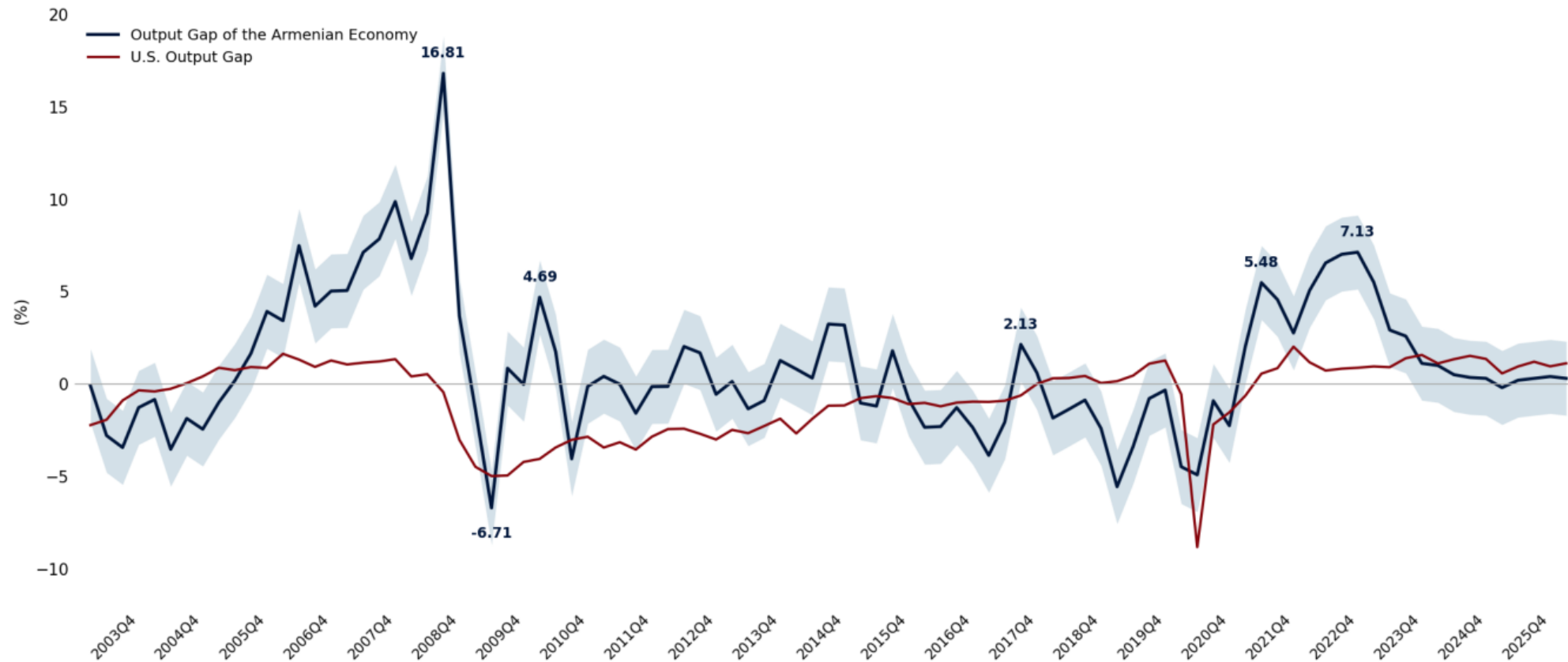
Dark Corner: Double-digit Unemployment, Deflation





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Figure 4: Historical output gaps, Armenia and U.S.



Source: Quarterly projection model for Armenia, Fred, CBO



The Loss Function

$$Loss_t = \sum_{j=0}^{\infty} \rho^j [\omega_1 (\pi_{t+j} - \pi^*)^2 + \omega_2 y_{t+j}^2 + \omega_3 (i_{t+j} - i_{t+j-1})^2] \quad (4)$$





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